

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER 2019-2020

MATHEMATICS FOR COMPUTER SCIENTISTS (COMP1017)

Time allowed ONE hour and THIRTY minutes only

Candidates must NOT start writing their answers until told to do so

There are THREE questions in the exam paper. Answer all THREE questions.

Only a calculator from approved list A may be used in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

Model Solutions

K – Knowledge
C – Comprehension
P – Problem solving

1. Questions on Logic, Sets, Functions, and Relations.

(a) For each statement, say whether it is **true** or **false**.

[5 marks]

(K, C)

(i) $(x \neq 3) \wedge (\exists x \in \mathbb{Z} \ x \geq 2)$ is a proposition. (false)(ii) Consider a relation R where $\exists a, b, c : (a, b), (b, c) \in R \not\Rightarrow (a, c) \in R$. R is not transitive. (true)(iii) Consider $y = \pm \sqrt{\frac{4-x}{a}}$. y is a function of x and is onto. (false)(iv) $\{0, 2, 12, 36, 80\} = \{y \in \mathbb{N} \mid \exists x \in \mathbb{N} ((y = x^2 + x^3) \wedge (0 \leq x \leq 5))\}$. (true)(v) Propositional operator $\vee$ is a function mapping propositions to propositions. (false)(b) Let $A = \{a, b, c, d, e\}$. Let S be the set of all functions on A , and T the set of all functions on the power set 2^A .(i) Provide an example of (x, y) for $y = g(x)$ where $g \in T$.

[2 marks]

(K)

For $y=g(x)$, $(\{a,b\}, \{d\})$ or any other valid example. [2 marks](ii) If all functions in S and T are total, determine $|S|$ and $|T|$. Explain your answer briefly.

[4 marks]

(K, C)

 $|S| = |A|^{|A|} = 5^5$; [2 marks] $|T| = (2^{|A|})^{(2^{|A|})} = (2^5)^{(2^5)} = 2^{160}$. [2 marks](iii) Let $f_1 = \{(a, b), (b, b), (c, e), (d, d), (e, b)\}$ and $f_2 = \{(a, d), (b, e), (e, c)\}$ where $f_1, f_2 \in S$. Determine $f_1 \circ f_2$ and explain whether it is one-to-one and whether it is onto.

[4 marks]

(C)

 $f_1 \circ f_2 = \{(a, d), (b, b), (e, e)\}$ [2 marks]

It is one-to-one [1 mark]

It is not onto. [1 mark]

- (iv) Define a function $g \in T$ such that the range of g is $\{\{d\}, \emptyset\}$, where $\emptyset$ is null set, with suitable set operator(s).

[3 marks]

(C, P)

We can use the set operator intersection [1 mark] to define a function g such that g takes as input any subset $B \subseteq A$ [1 mark] and intersect with set $\{d\}$ [1 mark], the result will be either $\{d\}$, or null set.

[This is to say that the model answer is $g(X) = X \cap \{d\}$ iff $X \subseteq A$ (or eqv. $X \in 2^A$). And they get 1 point for mentioning intersection, 1 for restricting g to the right domain, and 1 for taking the intersection with d .]

- (c) Let S be a finite set with n elements. Prove that the number of reflexive relations that can be defined on S is $2^{(n^2-n)}$. Show your steps.

[7 marks]

(C)

The cardinality $|S \times S| = n^2$ [1 mark]

A relation is a subset of $S \times S$ [1 mark]

A relation R on S is reflexive iff for all s in S , (s,s) is in R . [1 mark]

For any reflexive relation R on S we must have all n pairs of the form (s,s) [1 mark]

There are $n^2 - n$ pairs in $S \times S$ that does not fulfil the form of (s,s) [1 mark]

Each of these can be added in the relation R which still keep R reflexive. [1 mark]

Therefore the total number of reflexive relations on S is $2^{(n^2-n)}$. [1 mark]

Alternatively, one might answer that every pair either is or is not present in S except for the pairs (x,x) which must be present [3 marks]. So there are $n^2 - n$ elements that may or may not be in S , for a total of $2^{(n^2-n)}$ combinations. [4 marks]

Other alternative solutions will be awarded marks accordingly.

2. Questions on Number Theory, Matrix, and Mathematical Induction.

- (a) For each statement, say whether it is **true** or **false**.

[5 marks]

(K)

- (i) Let a and $b \geq 1$ be integers. There exist unique integers q and r such that $a = bq + r$.

(false)

(ii) Let A, B , and C be matrices. If $AC = BC$, then $A = B$.

(false)

(iii) Let $\{a_n\}, n \geq 1$ be a sequence of numbers, then $\forall k \in \mathbb{Z}, \sum_{n=5}^{10} a_n = \sum_{n=k-10}^{k-5} a_{k-n}$.

(true)

(iv) Let g be greatest common divisor of integers m, n . Let $g = 2^{a_1} \cdot 3^{a_2} \cdot 7^{a_3} \dots$ be written in its prime factorization form. If some $a_i = 1$ and all others are 0, then g is a prime number.

(true)

(v) $\gcd(252, 489) = \gcd(15, 10)$.

(false)

(b) Let p be a prime. For integers a, b , if $p|ab$, then $p|a$ or $p|b$.

(i) Using mathematical induction, prove that if $p|a_1 a_2 \dots a_n$, where each a_i is an integer, then $p|a_i$ for some i .

[7 marks]

(C, P)

Let $P(n)$: if $p|a_1 a_2 \dots a_n$, then $p|a_i$ for some i .

Basic step: prove that $P(1)$ holds. $P(1)$: if $p|a_1$, then $p|a_i$ for some i . Clearly, given that $p|a_1$, $p|a_i$ where $i=1$. [1 mark]

Inductive step: Assume that $P(k)$ holds, i.e., If $p|a_1 a_2 \dots a_k$, then $p|a_i$ for some i . We would like to prove that $P(k+1)$ also holds. [1 mark]

For $P(k+1)$, we have that $p|a_1 a_2 \dots a_k a_{k+1}$. [1 mark]

Let $b = a_k a_{k+1}$, then we have $p|a_1 a_2 \dots a_{k-1} b$. [1 mark]

According to $P(k)$, then $p|a_i$ for some i . [1 mark]

If $1 \leq i \leq k-1$, then we proved $P(k+1)$ holds. [1 mark]

If $p|a_i$ where $a_i = b = a_k a_{k+1}$, then $p|a_k$ or $p|a_{k+1}$, where $i=k$ or $i=k+1$. Thus we proved that $P(k+1)$. [1 mark]

Alternative solutions will be awarded marks accordingly. For example, The inductive step could start with the observation that if $p|a_1 \dots a_{k+1}$, then $p|a_1 \dots a_k$ or $p|a_{k+1}$, as per the statement in 2b. In case of the former, from the induction hypothesis, there is an $i \leq k$ s.t. $p|a_i$. In case of the latter, $p|a_{k+1}$.

(ii) Given that $p|a^9$, $p|(a^2 + b^2)$, show that $p|b^2$ based on (i).

[5 marks]

(C, P)

Since $p|a^9$ then $p|a$ (based on (i)) [1 mark]

$p|a$, then $p|a^2$ [1 mark]

Then $a^2 = mp$ for some integer m [1 mark]

Since $p|(a^2 + b^2)$, then $(a^2 + b^2) = np$ for some integer n [1 mark]

Thus $b^2 = np - a^2 = np - mp = (n - m)p$, [1 mark]

Since $n-m$ is an integer, thus $p|b^2$.

(c) Let $\{p_n\}, 1 \leq n \leq 9$ be the sequence of the first 9 prime numbers.

- (i) Store the sequence in a 3×3 matrix A by filling in each column from top to bottom and from the left most column to the right most column. Provide a general formula to determine the index (i, j) of an entry in A . What is the specific entry index that stores the prime number 17? Explain your answer.

[4 marks]

(C, P)

The first 9 prime numbers are $\{2, 3, 5, 7, 11, 13, 17, 19, 23\}$

To determine the row index i , we have $i = (n - 1) \bmod 3 + 1$ [1 mark]

To determine the column index j , we have $j = \lfloor n/3 \rfloor + 1$ [1 mark]

Recognising that prime 17 is the 7th number in the sequence and applying $n=7$ to the formula above we obtain that

$$i = (n - 1) \bmod 3 + 1 = 6 \bmod 3 + 1 = 0 + 1 = 1 \text{ [1 mark]}$$

$$j = \lfloor n/3 \rfloor + 1 = \lfloor 7/3 \rfloor + 1 = 2 + 1 = 3 \text{ [1 mark]}$$

- (ii) Let $C = AA^t$. Show that the sum of the diagonal entries of C is equivalent to $\sum_{n=1}^9 p_n^2$.

[4 marks]

(C)

$$A = \begin{bmatrix} P_1 & P_4 & P_7 \\ P_2 & P_5 & P_8 \\ P_3 & P_6 & P_9 \end{bmatrix} \text{ [0.5 mark]}$$

$$C = AA^t = \begin{bmatrix} P_1 & P_4 & P_7 \\ P_2 & P_5 & P_8 \\ P_3 & P_6 & P_9 \end{bmatrix} \begin{bmatrix} P_1 & P_2 & P_3 \\ P_4 & P_5 & P_6 \\ P_7 & P_8 & P_9 \end{bmatrix} \text{ [1 mark]}$$

$$\text{The diagonal entry of } C \text{ are } C_{1,1} = A_{1, \cdot} (A^t)_{\cdot, 1} = (P_1 \ P_4 \ P_7) \begin{pmatrix} P_1 \\ P_4 \\ P_7 \end{pmatrix} = P_1^2 + P_4^2 + P_7^2$$

$$\text{Similarly we have } C_{2,2} = P_2^2 + P_5^2 + P_8^2, \text{ and } C_{3,3} = P_3^2 + P_6^2 + P_9^2$$

[2 marks]

$$\text{Thus } C_{1,1} + C_{2,2} + C_{3,3} = \sum_{n=1}^9 p_n^2. \text{ [0.5 mark]}$$

3. Questions on Counting and Probabilities.

(a) For each statement, say whether it is **true** or **false**.

[5 marks]

(K, C)

- (i) Throw an unbiased coin 5 times and the probability of obtaining 3 heads is
- $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$
- .

(false)

- (ii) Put 4 apples into 2 boxes. The probability of one of the boxes having exactly 2 apples is 1.

(false)

- (iii) Hariz has either coffee or tea for breakfast but never both. He has coffee 40% of the time. Therefore, Hariz having coffee for breakfast is independent of him having tea for breakfast.

(false)

- (iv) The probability of a student studying hard but failing the subject is 0.1. The probability of the student failing the subject but not studying hard is 0.6. So the probability of the student failing the subject is 0.7.

(true)

- (v) Roll a dice of 6 sides with sides numbered from 1 to 6. The probability of observing a number that is greater than 0 is 1.

(true)

(b) There are 400 students in a programming class. On the midterm test each student received one of the following grades: A, B, C , or D .

- (i) Show that at least two of the students in the class were born on the same day of a month.

[4 marks]

(C, P)

Applying generalised pigeonhole principle, there are at least $\lceil 400/12 \rceil = 34$ students in one of the 12 months. [1.5 mark]

There are either 30 or 31 days in a month. [1 mark]

Applying pigeonhole principle, since there are at least 34 students in a month which is more than the total number of days (30 or 31) in a month, then there must be at least two students were born on the same day of a month. [1.5 marks]

[1 mark for recognising pigeons and pigeonholes in each case even if the final answer was

not correct]

Alternative solutions will be awarded marks accordingly. E.g., a student applies the pigeonhole principle to the whole year [2 marks] to figure out that two students must be born on the same day [2 marks]. Whatever that month is, is a month where two students were born on the same day.

- (ii) There are less students born in the first half of a year than in the latter. 60% of the grade A students were born in the first half of a year. Under what circumstances can one say that the event of students born in the first half of a year is independent of the event of students obtaining grade A. Explain your answer.

[4 marks]

(C)

Let E be the event that student was born in the first half of a year.

Let F be the event that student obtained grade A.

We have that $P(E) < 0.5$ [0.5 marks],

$P(E, F) = 0.6$ [0.5 marks]

To check if E and F are independent of each other, we would like to see if $P(E, F) = P(E)P(F)$ [0.5 marks]

Assume E and F are independent, then we have $P(F) = P(E, F)/P(E) = 0.6/P(E)$ [1 mark]

Since $P(E) < 0.5$, then $1/P(E) > 1/0.5 = 2$, then $P(F) = 0.6/P(E) > 0.6 * 2 = 1.2$ which is not a valid probability. [1 mark]

Therefore, under no circumstances that the two events are independent. [0.5 marks]

(c) A bag contains 6 red, 2 blue, and 5 yellow marbles.

(i) In how many ways can one select 5 marbles.

[4 marks]

(K, C)

Recognise that the problem is combination with repetitions allowed.

However, it is not possible to have more than 2 blue marbles in any of the combination, we then have to use the selection of blue marbles to categorise the cases, i.e., 1) no blue marbles, 2) 1 blue marble, or 3) 2 blue marbles.

For 1), we need to select 5 marbles from 2 categories. So it's $C(5+2-1,1)$ [1 mark]

For 2), we need to select 4 marbles from 2 categories, so it's $C(4+2-1,1)$ [1 mark]

For 3), we need to select 3 marbles from 2 categories, so it's $C(3+2-1,1)$ [1 mark]

So the final answer is $C(6,1)+C(5,1)+C(4,1)$ [1 mark]

[1 mark for recognising it's combination with repetitions allowed if no solutions are provided. Full marks will be given for any other valid solutions]

(ii) A marble is selected at random and its colour is noted. If the marble is blue, it is put back in the bag, otherwise it is not. A second marble is drawn and its colour is noted. Find the probability that the first marble is blue, given that the second marble is yellow.

[8 marks]

(C, P)

Let R, B, Y represent the marble being red, blue, and yellow, respectively.

For the first marble selected, we have

$$P(1^{\text{st}}=R)=6/(6+2+5)=6/13 \text{ [0.5 marks]}$$

$$P(1^{\text{st}}=B) = 2/13 \text{ [0.5 marks]}$$

$$P(1^{\text{st}}=Y) = 5/13 \text{ [0.5 marks]}$$

For the second marble selected, we have

$$P(2^{\text{nd}}=R|1^{\text{st}}=R) = (6-1)/12 = 5/12 \text{ [0.5 marks]}$$

$$P(2^{\text{nd}}=B|1^{\text{st}}=R) = 2/12 = 1/6 \text{ [0.5 marks]}$$

$$P(2^{\text{nd}}=Y|1^{\text{st}}=R) = 5/12 \text{ [0.5 marks]}$$

$$P(2^{\text{nd}}=R|1^{\text{st}}=B) = 6/13 \text{ [0.5 marks]}$$

$$P(2^{\text{nd}}=B|1^{\text{st}}=B) = 2/13 \text{ [0.5 marks]}$$

$$P(2^{\text{nd}}=Y|1^{\text{st}}=B) = 5/13 \text{ [0.5 marks]}$$

$$P(2^{\text{nd}}=R|1^{\text{st}}=Y) = 6/12 \text{ [0.5 marks]}$$

$$P(2^{\text{nd}}=B|1^{\text{st}}=Y) = 2/12 = 1/6 \text{ [0.5 marks]}$$

$$P(2^{\text{nd}} = Y | 1^{\text{st}} = Y) = (5-1)/12 = 1/3 \text{ [0.5 marks]}$$

We'd like to calculate $P(1^{\text{st}} = B | 2^{\text{nd}} = Y)$. Applying Bayes Theorem, we have

$$P(1^{\text{st}} = B | 2^{\text{nd}} = Y) = (P(2^{\text{nd}} = Y | 1^{\text{st}} = B)P(1^{\text{st}} = B)) / (P(2^{\text{nd}} = Y | 1^{\text{st}} = R)P(1^{\text{st}} = R) + P(2^{\text{nd}} = Y | 1^{\text{st}} = B)P(1^{\text{st}} = B) + P(2^{\text{nd}} = Y | 1^{\text{st}} = Y)P(1^{\text{st}} = Y)) \quad [1 \text{ mark}]$$

$$= (5/13 * 2/13) / (5/12 * 6/13 + 5/13 * 2/13 + 1/3 * 5/13) \quad [1 \text{ mark}]$$

$$= 0.1558$$

[full marks will be awarded for any other valid solutions, e.g., based on tree diagram.]